Final Project - Reliability in Machine Learning Risk-Averse Calibration (RAC) for Decision-Making, with Group-Conditional and Online Extensions

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GitHub: https://github.com/husseinrayan1/risk_averse_conformal

1 Abstract

This project studies Risk-Averse Calibration (RAC) for conformal prediction in downstream decision-making, as introduced in “Decision Theoretic Foundations for Conformal Prediction: Optimal Uncertainty Quantification for Risk-Averse Agents”. RAC connects prediction sets to risk-averse decision policies via a max-min objective, targeting high utility while controlling a user-specified risk level (miscoverage) α.

We implement RAC in a recommendation-style classification setting (MovieLens 100K), using a learned probabilistic predictor over discrete outcomes (ratings). Beyond reproducing a working global RAC pipeline, we extend it in three directions motivated by reliability under heterogeneity and non-stationarity: (i) group-conditional RAC for demographic groups (gender), (ii) adaptive group discovery via calibration-time clustering to define groups where reliability issues emerge, and (iii) an online RAC extension that recalibrates the RAC parameter over a rolling buffer, enabling tracking and adaptation under synthetic distribution drift.

Empirically, we observe the core reliability-utility trade-off of RAC across α and show how group conditioning and adaptive grouping provide additional diagnostic power and can reduce worst-group failures, at the cost of increased conservativeness. In the online setting, the controller adapts its calibrated parameter trajectory following distribution shifts, with corresponding changes in miscoverage, utility, and set size.

2 Background and problem setup

In many machine learning applications, predictions are not consumed directly; rather, they inform actions taken by an agent. When predictions are uncertain, a purely expected-utility policy can perform poorly in adverse outcomes, while a fully conservative policy can sacrifice too much utility. Reliability in this context means producing uncertainty information aligned with risk-sensitive decision-making.

Conformal prediction provides distribution-free guarantees for prediction sets under exchangeability, typically expressed as a miscoverage constraint.

$$P(Y \notin C(X)) \leq \alpha.$$

RAC builds on this to optimize decisions for agents who care about Value-at-Risk (VaR)-type objectives, resulting in decision rules tied to max-min utility.

3 Decision Theoretic Foundations for Conformal Prediction

This section describes the baseline algorithmic components as implemented in the codebase and notebooks, and the quantities used for evaluation.

3.1 Predictive model (MovieLens classifier)

We train a neural classifier on MovieLens 100K. The dataset is built from data (ratings), user demographics, and items (movie genres/features). We preprocess into a classification dataset and split randomly into train, calibration, and test using a fixed seed.

3.2 RAC Decision Mechanism

For a test point x, the model outputs predicted probabilities over outcomes $y \in \mathcal{Y}$. Given an action set $\mathcal{A}$and utility function $u(a, y)$, RAC constructs decisions in two steps.

First, fix a probability level $t \in [0,1]$.
For each action a, define the total probability of outcomes achieving utility at least u as

$T(a, u)$ = sum of predicted probabilities of all y such that $u(a, y) \geq u$.

The largest utility that action a can guarantee with probability at least t is the largest u such as $T(a, u) \ \geq \ t$.
Among all actions, we choose the one with the highest guaranteed utility at level t.
Denote this value by $h(x, t)$.

Second, RAC selects the probability level s by maximizing

$$\beta \ \cdot \ s \ + \ h(x, s),$$

where $\beta \geq 0$ is the calibrated parameter controlling the safety–utility tradeoff.

The final prediction set contains all outcomes whose utility under the selected action is at least the guaranteed utility level. Coverage is the fraction of test examples whose true outcome lies in this set.

3.3 Calibration procedure for beta (global RAC)

The calibration parameter β is chosen using a held-out calibration set.

For a given candidate value of β, we:

1. Construct a prediction set for each calibration example using the procedure above.
2. Measure the fraction of calibration points whose true outcome lies inside their prediction set.
3. Compute empirical coverage.

We want coverage to be at least $1-\alpha$.

To find such a β We use a monotone search procedure:

• First, increase β until coverage becomes large enough.
• Then perform a binary search between feasible and infeasible values.

The final value of $\beta$is the smallest one that achieves the desired coverage level.

3.4 Evaluation metrics (offline)

For each α in {0.005, 0.01, 0.02, 0.05, 0.1, 0.15, 0.2, 0.25}, the test evaluation computes:
• Coverage and miscoverage
• Average realized utility and average max-min value
• Utility α-quantile
• Action counts and diagnostics (utility histogram/PDF)
These metrics are computed exactly as in the notebooks.

4 Extensions

4.1 Group-Conditional RAC (Gender groups)

Group definition: groups are defined by user gender. In preprocessing, gender is encoded as $M \to 0, F \to 1$. Per-group calibration: a separate scalar threshold β_g is calibrated for each group using the same doubling and binary search structure but computed on group-restricted calibration data.

Formally, let

$$I_g = \{ i : g_i = g \}$$

denote the calibration indices belonging to group g.

For each group g, We select the smallest β_g such that

$$\left(\frac{1}{|I_g|}\right) \sum_{\{i \in I_g\}} 1\{ y_i \in C(x_i; \beta_g) \} \geq 1-\alpha$$

This enforces coverage constraints separately within each group.

4.2 Adaptive group discovery via clustering

To move beyond predefined demographic splits, we introduce an adaptive grouping strategy based on clustering. Instead of fixing groups according to observable attributes (such as gender), we allow the data itself to define subpopulations where reliability behavior may differ.

Specifically, we apply a clustering algorithm to feature representations of the calibration set only. The clustering model is trained exclusively in calibration data to avoid any information leakage from the test set. Once fitted, the clustering model is frozen and used to assign both calibration and test samples to cluster identities.

These cluster assignments serve as adaptive group labels. We then apply the same group-conditional RAC procedure, calibrating a separate parameter for each discovered cluster and evaluating coverage and utility per cluster. This allows us to detect and control reliability disparities that may arise in regions of the feature space not captured by simple demographic splits.

4.3 Online RAC extension (streaming recalibration)

To address non-stationarity, we extend RAC to a streaming setting in which data arrives sequentially, and the underlying distribution may change over time.

Instead of performing calibration once on a fixed dataset, we process samples in chronological order. At each time step t, the algorithm receives a new input x_t, constructs a prediction set using the current calibration parameter $\beta(t)$, selects an action, observes the true outcome y_t, and records coverage and realized utility.

To enable adaptation, we maintain a rolling calibration buffer consisting of the most recent observations. At regular intervals, the calibration parameter $\beta(t)$is recomputed using only the data currently stored in this buffer. In this way, the system continuously updates its level of conservativeness based on recent performance rather than relying on a fixed calibration performed in the past.

This online mechanism allows RAC to respond dynamically to changes in prediction quality or data distribution.

We evaluate the online extension under two scenarios:

1. **Stationary**: the data-generating process remains unchanged over time.
2. **Drift**: at a predefined time point, prediction quality is degraded to simulate a distribution shift.

To simulate drift, after the shift time, we corrupt the model's predicted probability vectors by mixing them with the uniform distribution. This reduces confidence and effectively worsens predictive accuracy, mimicking a real-world distribution change.

We monitor the following time-dependent quantities:

- Rolling miscoverage
- The trajectory of the calibrated parameter $\beta(t)$
- Rolling average realized utility
- Rolling prediction set size

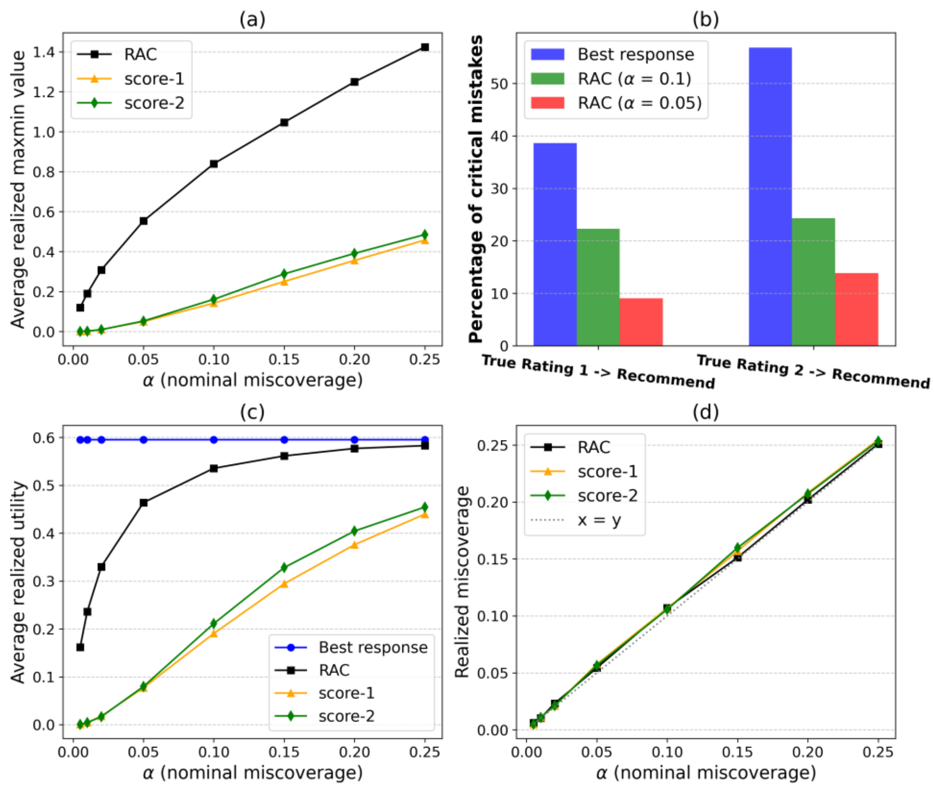
In the stationary regime, $\beta(t)$stabilizes, and coverage remains close to the target level. Under drift, miscoverage initially increases; the recalibration mechanism then adjusts $\beta(t)$, leading to larger sets and recovery of coverage.

This extension demonstrates that RAC can be embedded in a sequential decision system and can adapt its level of conservativeness in response to distributional changes

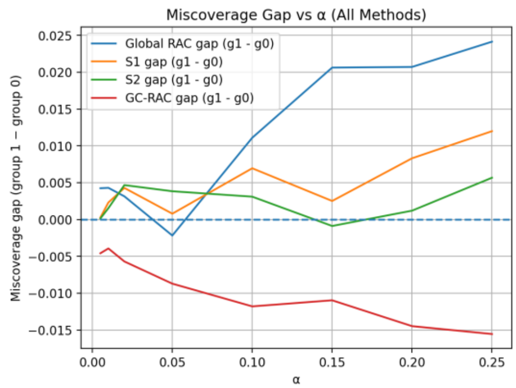
5 Results

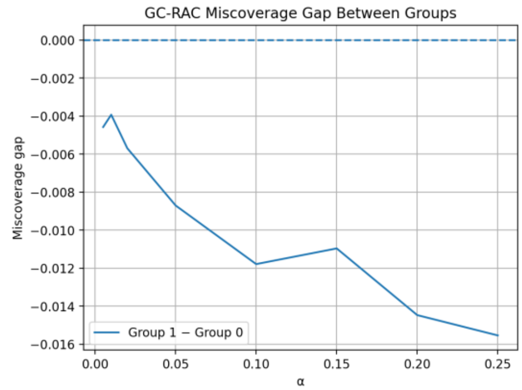
All figures below reference actual filenames from the results folder. Figures are inserted as place holders for easy replacement.

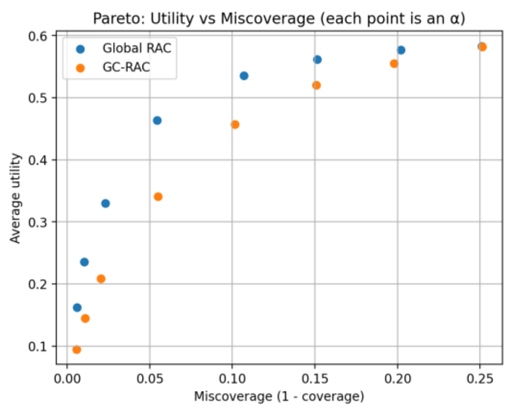
5.1 Global RAC: offline

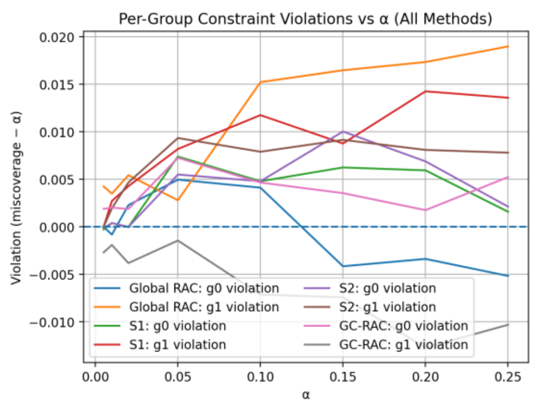


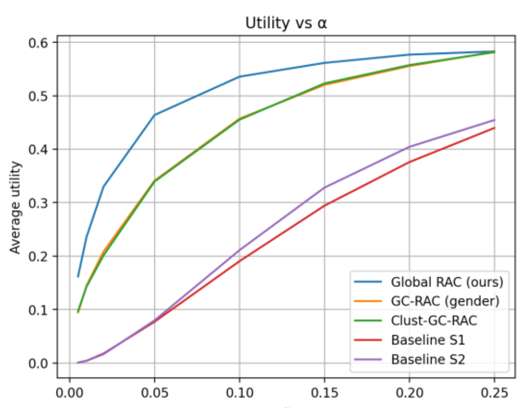
5.2 Group-Conditional RAC on gender

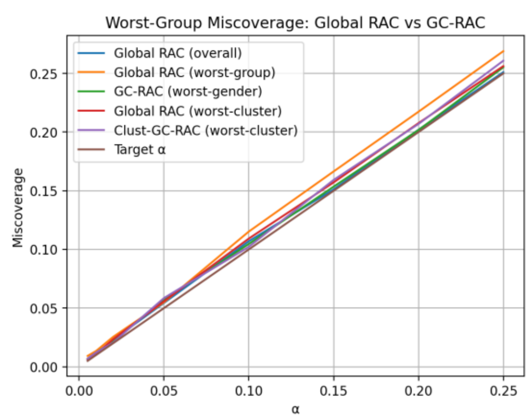






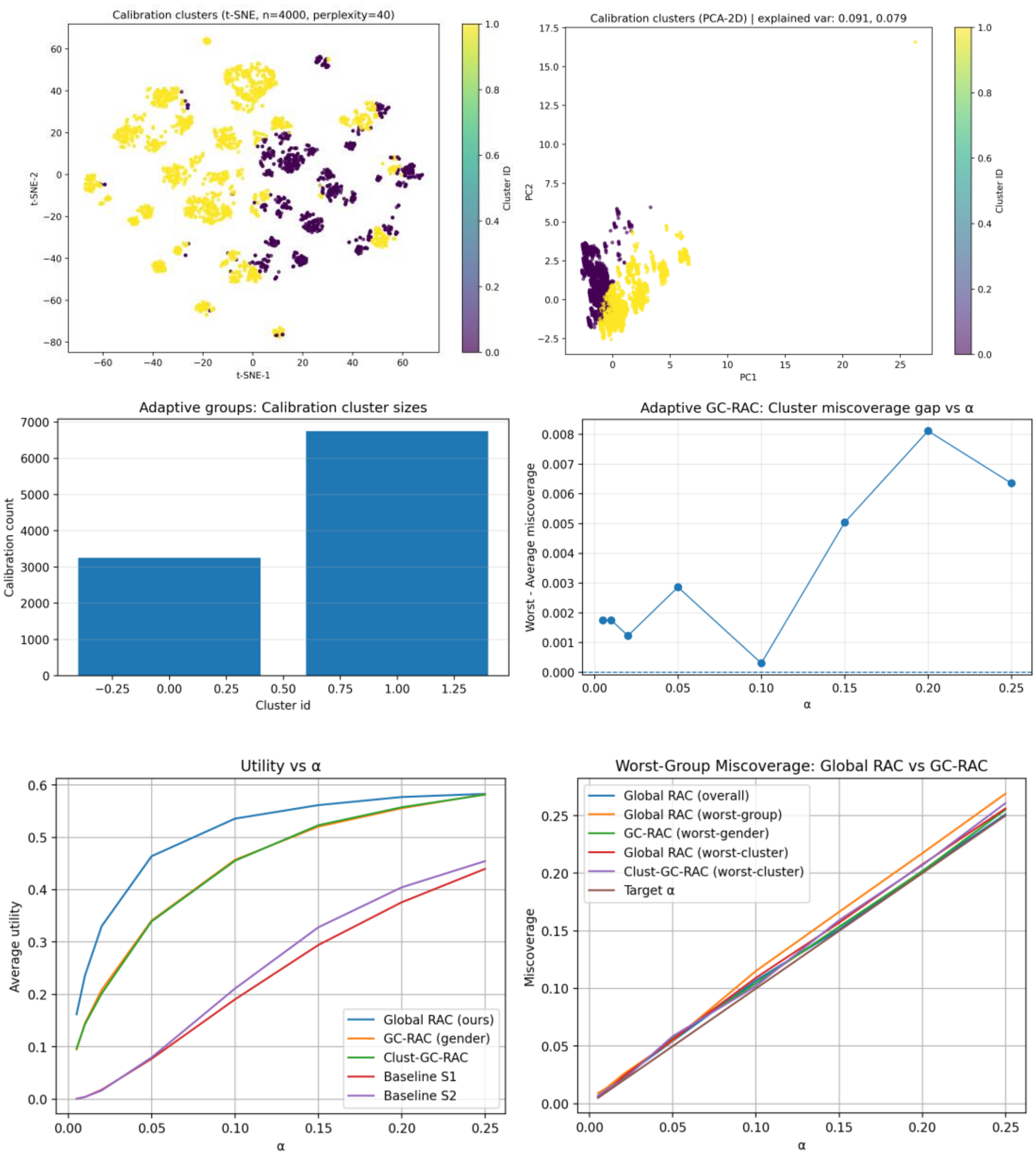






This experiment examines whether global calibration distributes reliability evenly across demographic subgroups. We compare global RAC with group-conditional RAC, where each gender group receives its own calibrated parameter. For each α, we evaluate overall miscoverage, per-group miscoverage, worst-group miscoverage, and the miscoverage gap between groups. We also compare average realized utility and average max–min value. While global RAC controls marginal miscoverage, group-conditional calibration reduces worst-group violations by enforcing coverage separately within each group. This typically results in slightly lower average realized utility, reflecting the cost of stronger subgroup reliability guarantees.

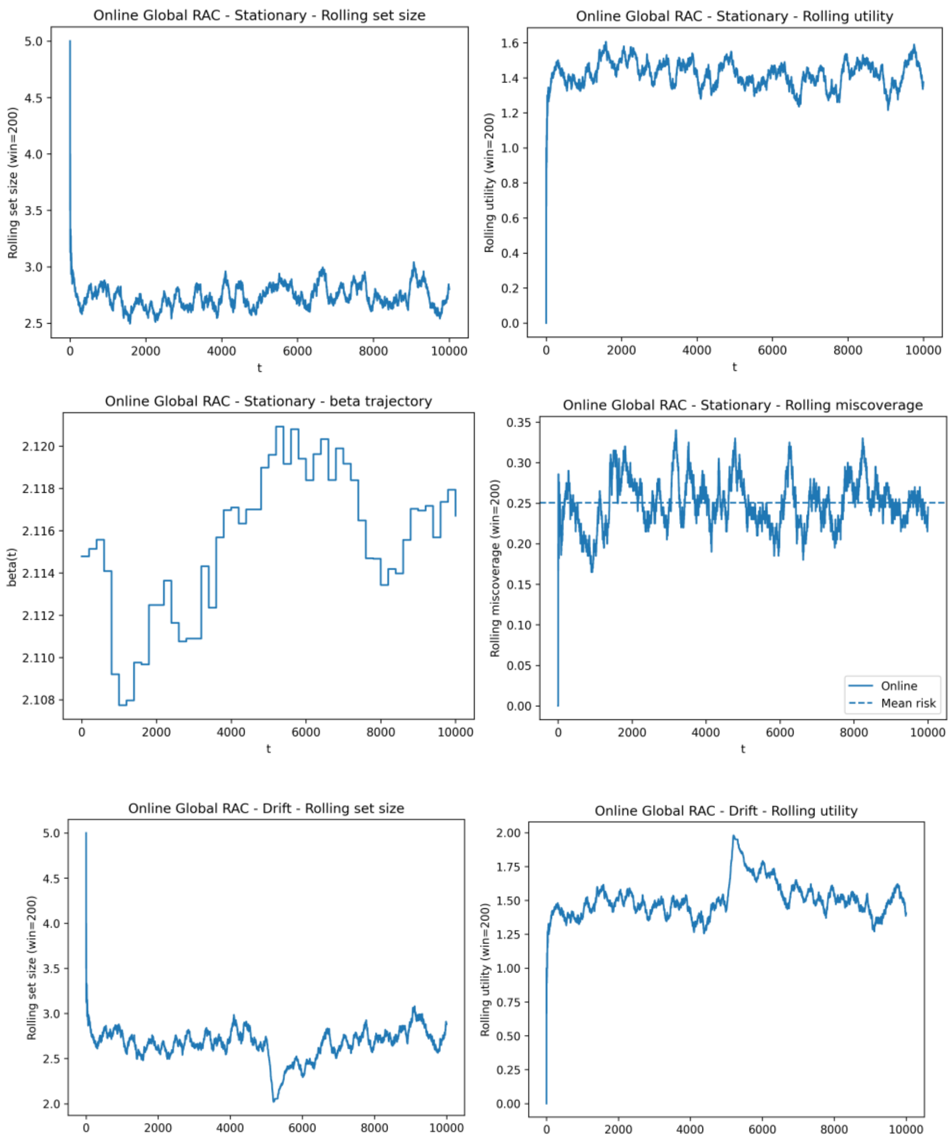
5.3 Adaptive group-conditional RAC via clustering

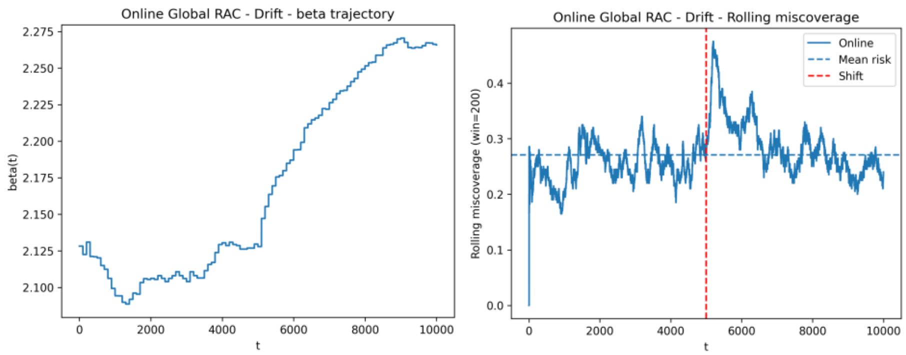


To identify reliability disparities not captured by demographic attributes, we construct adaptive groups using clustering on calibration features. Each cluster is treated as a separate subgroup and calibrated independently.

We evaluate per-cluster miscoverage, worst-cluster miscoverage, and cluster-level miscoverage gaps across α. In addition, we compare average realized utility and guaranteed utility to the global baseline. The results reveal that reliability failures can concentrate in specific regions of feature space. Adaptive grouping reduces worst-cluster miscoverage relative to global RAC, though at the cost of increased conservativeness in smaller clusters.

5.4 Online global RAC: stationary vs drift





In the online setting, we evaluate RAC under sequential data arrival. At each time step, we record realized utility, max–min value, miscoverage, and prediction set size using a rolling window.

Under stationary conditions, rolling miscoverage remains close to the target level, and the calibrated parameter stabilizes, indicating consistent reliability over time. Under synthetic drift, miscoverage initially increases due to degraded predictive quality. The recalibration mechanism then adjusts the parameter upward, enlarging prediction sets and restoring coverage. This demonstrates that online RAC can dynamically adapt its conservativeness to maintain reliability under distribution shifts.

6 Conclusions

In this project, we implemented and empirically evaluated Risk-Averse Calibration (RAC) in a discrete-outcome recommendation setting based on MovieLens 100K. The global RAC baseline successfully enforces the desired miscoverage level across a range of α values, confirming that the calibration procedure effectively controls marginal reliability. As expected, increasing α relaxes the reliability constraint, leading to higher realized utility and smaller prediction sets, illustrating the fundamental reliability–utility tradeoff inherent in risk-aware decision-making.

However, our experiments show that marginal guarantees alone are insufficient to ensure uniform reliability across subpopulations. Under global calibration, certain groups can experience higher miscoverage than the nominal level. Introducing group-conditional RAC reduces worst-group violations and narrows miscoverage gaps, at the cost of modest reductions in average realized utility. This highlights a key tradeoff between subgroup fairness and global efficiency.

The adaptive clustering extension further demonstrates that reliability disparities can arise in regions of feature space not captured by simple demographic attributes. By calibrating per cluster, we reduce worst-cluster miscoverage and gain finer-grained diagnostic insight into model behavior. These results emphasize that heterogeneity in reliability may be structural rather than purely demographic.

In the streaming setting, the online RAC extension remains stable under stationary conditions and successfully adapts under a synthetic distribution shift. Following drift, rolling miscoverage increases temporarily, after which the recalibration mechanism adjusts the conservativeness parameter, enlarges prediction sets, and restores coverage. This demonstrates that RAC can be embedded in sequential systems and can dynamically respond to non-stationarity.

Overall, our findings reinforce the central insight of RAC: reliability control fundamentally interacts with decision quality. Enforcing stronger guarantees improves safety but increases conservativeness. Extensions that account for subgroup structure and temporal dynamics improve robustness but introduce additional complexity and tradeoffs.

7 Future Work

Stronger integration of group-conditional RAC with the baseline objective.
In the current implementation, the group-specific parameter β_g is calibrated independently

within each group and then used directly for set construction. A more principled extension would integrate group-conditional calibration fully into the original RAC objective, preserving the same feasibility–utility optimization structure while enforcing per-group constraints. This would provide a cleaner theoretical alignment between the global and group-specific formulations.

Statistical stability under small or data-driven groups.
Adaptive clustering can generate small clusters, which may lead to unstable calibration due to limited sample size. Future work could incorporate minimum cluster size constraints, cluster merging procedures, or hierarchical grouping strategies. Regularized or partially pooled calibration across related groups may also improve stability while maintaining subgroup reliability guarantees.

Online group-conditional RAC under heterogeneous drift.
In our current experiments, drift affects the full population uniformly. A more realistic setting involves group-specific shifts, where only certain subpopulations experience degradation. Evaluating online group-conditional RAC under such heterogeneous drift would test whether per-group recalibration can prevent worst-group reliability failures more effectively than a single global parameter.

Extension to sequential decision-making problems.
The present work considers one-shot decisions: for each input, a single action is selected based on a calibrated uncertainty set. However, many real-world systems operate in sequential settings, where actions influence future states and decisions are taken repeatedly over time. Extending RAC to sequential decision problems, such as reinforcement learning or dynamic control, would require integrating conformal uncertainty with multi-step planning under uncertainty. This introduces significantly greater complexity, as reliability guarantees must account for temporal dependence and cumulative risk. Developing risk-aware conformal methods for sequential decision-making remains an important and challenging direction for future research.

References

Kiyani, S., Pappas, G., Roth, A., Hassani, H. Decision Theoretic Foundations for Conformal Prediction: Optimal Uncertainty Quantification for Risk-Averse Agents.
GroupLens Research. MovieLens 100K Dataset.